MIL-G-46884(MI)
14 December 1973
SUPERSEDING
MIS-14362
15 November 1965

MILITARY SPECIFICATION

GLASS, METAL SEALING

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers one type of glass intended for use where a coefficient of thermal expansion approximating that of cobalt-iron-nickel alloy is desired.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-G-174	Glass, Optical

STANDARDS

Federal

FED-STD-102	Preservation, Packaging, and Packing Levels
FED-STD-406	Plastics, Methods of Testing

FSC 8030

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1167	Ammunition Data Card

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM C 336	Tentative Test Method for Annealing Point and Strain Point of Glass
ASTM E 228	Standard Method of Test for Average Linear Expansion of Glass
ASTM E 111	Test for Young's Modulus at Room Temperature

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (See 6.2), a preproduction sample of glass shall meet the requirements of this specification.

3.2 Material. The glass shall be fine annealed, clear, glass formulated from silica, boric oxide, soda, and alumina oxides.

3.3 Mechanical properties.

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3.3.1 Coefficient of expansion. The coefficient of expansion shall not be greater than 0.0000046 inch per inch per degree celsius (C) (1.8 degrees Fahrenheit (F) in the range of zero degrees C to 300 degrees C (32 degrees F to 572 degrees F)).

3.3.2 Modulus of elasticity. The modulus of elasticity shall be not less than 8.2 times 10^6 pounds per square inch.

3.4 Physical properties.

3.4.1 Specific gravity. The specific gravity shall be 2.280 plus or minus 0.004.

3.4.2 Strain point. The strain point shall be not greater than 435 degrees C (815 degrees F).

3.4.3 Annealing point. The annealing point shall be not greater than 480 degrees C (896 degrees F).

3.5 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform, and in conformance with this specification. The glass shall be free of dirt, foreign materials, cracks, inclusions or contaminants.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of glass. Preproduction samples which do not meet the requirements of this specification shall be subject to rejection and return to the supplier.

4.3 Classification of examinations and tests. The examination and testing of the glass shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

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4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the glass shall consist of the following examinations and tests:

a. Specific gravity.

b. Workmanship.

4.3.2.1 Lot size. A lot shall consist of all the glass submitted for acceptance by one manufacturer, which has been prepared without change in material or processes in one continuous period of operation.

4.3.2.2 Sampling. Unless otherwise specified (See 6.2C), fifty formed blanks shall be randomly selected as the representative sample of each lot for compliance to the quality conformance tests of 4.4 MIL-STD-105 , using code letter H with an AQL of one percent, shall apply.

4.4 Test methods.

4.4.1 Test conditions. Unless otherwise specified in the applicable test, all tests shall be conducted at 23 plus or minus 2 degrees C (73.4 plus or minus 3.6 degrees F) and at a relative humidity of 50 plus or minus 5 percent.

4.4.2 Mechanical properties tests.

4.4.2.1 Coefficient of expansion. The coefficient of expansion shall be determined in accordance with ASTM E 228. The coefficient of expansion shall be as specified in 3.3.1.

4.4.2.2 Modulus of elasticity. The modulus of elasticity shall be determined in accordance with ASTM E 111. The modulus of elasticity shall be determined as specified in 3.3.2.

4.4.3 Physical properties tests.

4.4.3.1 Specific gravity. The specific gravity shall be determined in accordance with FED-STD-406, Method 5011. The specific gravity shall be as specified in 3.4.1.

4.4.3.2 Strain point. The strain point shall be determined in accordance with ASTM C 336. The strain point shall be as specified in 3.4.2.

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4.4.3.3 Annealing point. The annealing point shall be determined in accordance with ASTM C 336. The annealing point shall be as specified in 3.4.3.

4.4.4 Workmanship. Examination for workmanship shall be in accordance with the examination for Inclusions, Method 1 of MIL-G-174. Workmanship shall be as specified in 3.5.

4.5 Rejection. Lots which do not meet the requirements of this specification shall be rejected and returned to the supplier.

4.6 Preservation, packaging, packing and marking. The preservation, packaging, packing and marking shall be examined for compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Unit packaging. Unless otherwise specified in the contract or purchase order, the formed blanks shall be packaged in accordance with Method III of MIL-P-116.

5.1.2 Intermediate packaging. When required (See 6.2C), specified quantities of unit packages shall be intermediately packaged as specified in the contract or purchase order.

5.1.3 Packaging level. Unit and intermediate packaging shall be in accordance with Level C of FED-STD-102 .

5.2 Packing. Packing shall be in accordance with Level C of FED-STD-102 .

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 .

5.3.1 Special marking. Shipping containers shall be marked with the words "GLASS - HANDLE WITH CARE." The appropriate side of the container shall be clearly marked to indicate "TOP" or "OPEN THIS SIDE."

6. NOTES

6.1 Intended use. The glass covered by this specification is intended for a glass to cobalt-iron-nickel alloy seal.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (See 3.1) and, if so, pertinent details.

c. Sampling plan if other than specified (See 4.3.2.2).

d. Any special marking requirements (See 5.3).

e. Packaging limitations (See 5.1).

f. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167 .

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14362, dated 15 November 1965.

Custodian:
ARMY-MI

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